

Simulation-informed transfer learning improves low-data nanobody thermal stability prediction

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Nanobodies are single-domain antibody fragments whose practical development often depends on thermal stability. A common experimental readout is the melting temperature (T_m), the temperature at which a protein loses half of its folded population under a given assay condition. Experimental T_m measurements are scarce for nanobodies, whereas molecular simulations and structure-based calculations can produce many related labels for mutated or designed sequences. The practical problem is that these computational labels are usually not T_m measurements. We therefore asked how such labels should be used for experimental T_m prediction when they measure related but different physical quantities. Using a public nanobody T_m benchmark, we trained transfer-learning models in which a pretrained protein sequence model learned the experimental T_m task together with one computational prediction task at a time. To keep the comparison experimental, checkpoint selection, model selection, and final evaluation were all based only on experimental T_m validation or test examples. We compared mutation free-energy changes from alchemical free-energy perturbation (FEP) calculations, structure-based mutation scores, learned stability-change scores, physics scores on generated and random sequence variants, and a molecular-dynamics trajectory summary that measures retained native contacts. The FEP mutation free-energy labels gave the clearest improvement: held-out T_m mean absolute error decreased from 6.61 to 6.26°C relative to training on T_m labels alone, with a paired improvement of 0.35°C (90% confidence interval, 0.24–0.46°C). The improvement was not explained by using a larger protein sequence model and was not reproduced by the contact-persistence trajectory summary. Structure-scored generated variants gave a weaker but positive signal. These results show that simulation data can improve low-data T_m prediction, but only when the computed label carries physical information that transfers to the experimental endpoint. In this case, sparse absolute T_m measurements and relative mutation free-energy changes provide complementary views of nanobody stability.

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Introduction

Simulation data for low-data experimental prediction

Machine learning has made it possible to screen large molecular and materials spaces before costly experimental testing, but its usefulness still depends on

the availability of reliable property labels. In materials informatics, this data bottleneck has motivated a simulation-to-real (Sim2Real) strategy: large computational databases generated by first-principles calculations, quantum chemistry, or molecular dynamics simulations are used as source data, and the resulting models are transferred to smaller experimental target datasets (1–3). A recent study in polymers and inorganic materials showed that the downstream prediction error of transferred models can scale with the amount of computational data, providing a quantitative way to judge whether a growing simulation database has real experimental value (3). This framing is important because simulation data are not useful by default: simulations and experiments are separated by model assumptions, sampling limits, and systematic domain gaps.

Protein engineering faces an analogous problem, but with an additional layer of physical mismatch. Experimental measurements of protein properties are expensive and sparse, whereas biomolecular simulations and physics-based calculations can generate labels for many designed or mutated sequences. The challenge is sharper than in many property-matched transfer settings, because the simulated quantity may not be the same as the experimental target. For example, a target phenotype may be the melting temperature (T_m), the temperature at which a protein loses half of its folded population under a given assay condition. A tractable calculation, by contrast, may estimate how much a mutation changes stability, assign a structure-based stability score, or summarize a molecular-dynamics trajectory. The central question is therefore not simply whether simulation data can be added to a training set, but which simulated physical quantities can transfer to the experimental phenotype.

Nanobody thermostability in the high-throughput era

We examine this question using nanobody thermostability as a concrete case. Nanobodies, or single-domain VHH antibodies, are attractive engineering scaffolds because of their small size, modularity, and binding properties (4, 5). Their practical use, however, depends on thermal stability during expression, storage, formulation, and downstream screening. Experimental T_m measurements remain limited, and direct molecular simulation of thermal stability is computationally demanding

(6). Sequence-based predictors and protein language models (PLMs) offer an attractive alternative: PLMs such as ESM2 learn broadly useful protein representations from large sequence corpora (7, 8), and recent methods have used deep sequence representations to predict protein or nanobody melting temperatures (9–11). High-throughput stability measurements are also changing the data landscape. Tsuboyama et al. introduced a cDNA-display proteolysis assay and used it to produce a mega-scale folding-stability dataset with approximately 776,000 high-quality measurements from natural and designed protein domains (12). That advance shows that experimental ΔG can now be measured at unprecedented scale for suitable small domains. However, the curated measurements span 40–72 amino-acid domains, whereas nanobody VHH domains are typically much longer, and comparable high-throughput ΔG measurements are not yet available for full-length nanobody VHH domains. A model fine-tuned on the mega-scale dataset also showed limited transfer to larger protein scaffolds: although it performed well on small protein domains, benchmarks on 177–501-residue proteins were substantially weaker than structure-based methods (13). Thus, large-scale ΔG data are powerful but do not remove the need for target-specific supervision and careful validation in nanobody Tm prediction (14–16).

Study design and main finding

Here we ask whether simulation-derived labels can provide that additional supervision when experimental Tm labels are scarce. We treat Tm prediction as the experimental target task and treat each computational label as a source task that can shape the learned sequence representation. The implementation uses ESM2, a pretrained protein language model, as the shared sequence encoder (8). The computational labels represent several kinds of computational information: mutation free-energy changes from alchemical free-energy perturbation (FEP) calculations, structure-based mutation scores from Rosetta, learned stability-change scores from ThermoMPNN, Rosetta scores on generated and random variants, and a molecular-dynamics (MD) trajectory summary measuring retained native contacts. To keep the comparison target-centered, all model settings are selected using an experimental validation set, and all final claims are evaluated on the same held-out experimental Tm test set with paired error comparisons. This design prevents high performance on a computational task from being mistaken for improved Tm prediction. The main result is that the benefit of simulation-derived supervision is highly source-dependent. Mutation free-energy labels from FEP provide the strongest improvement over training on Tm labels alone, whereas the MD contact-persistence summary does not improve held-out Tm prediction under the same evaluation design. Rosetta-scored generated variants provide a weaker but positive signal, suggesting a connection to future

generator–physics–predictor design loops. Taken together, these results extend the Sim2Real argument from materials informatics to a biomolecular setting in which the source and target labels differ in physical meaning. Useful simulation data are not merely abundant; they encode a physical quantity that transfers to the experimental property of interest.

Materials and Methods

Experimental and technical design

This study was designed to test whether computational source labels can improve experimental nanobody melting-temperature (Tm) prediction when the source label is physically related to, but not identical with, Tm. We therefore used a multi-task transfer-learning design: experimental Tm prediction was the target task, and computational labels were source tasks that shared part of the neural network with the target task. The central constraint was that all model selection and all final claims were anchored to experimental Tm performance. Source-task losses were used for representation learning during training, but source-task validation errors were not used to select final checkpoints or final hyperparameters.

Experimental Tm data set and split definitions

Experimental Tm labels were taken from a public nanobody benchmark data set, NbBench, which provides a thermo-seq task with predefined splits (15). The processed files used in this study contained 57 training sequences, 114 validation sequences, and 396 test sequences. The terms were used as follows. The training split was used to fit model parameters. The validation split was used for checkpoint selection, early stopping, and hyperparameter selection. The test split was held out until the selected settings were fixed and was used only for final reporting. This definition applies to both Tm-only models and multi-task transfer models.

The target CSV files contained an amino-acid sequence column and a Tm label in degrees Celsius. During training, Tm labels were transformed to a scaled regression target. The scaler was fit only on the 57 training labels, using a robust scaling step followed by min–max scaling to [0,1]. The fitted scaler was then applied to validation labels during training. At evaluation time, model predictions were inverse-transformed back to degrees Celsius before computing errors. The validation and test label distributions were therefore not used to fit the target-label scaling transform.

For the experimental-label scaling reference, the Tm training split was subsampled to the requested number of labels using the run seed. The validation and test splits were not subsampled. These experiments provide a target-label reference for interpreting whether computational source-label scaling behaves in the expected direction.

Computational source-label data sets

Computational labels were represented as source tasks rather than as additional Tm measurements. The source-label collection comprised mutation free-energy changes from alchemical free-energy perturbation (FEP), structure-based mutation-effect scores from Rosetta, learned mutation-effect scores from ThermoMPNN, Rosetta scores assigned to generated and random variants, and a molecular-dynamics (MD) contact-persistence label. All source labels were preprocessed to [0,1]-scaled regression targets and were used only through their own source-task heads.

Mutation-effect sources were organized as two template-specific tables derived from the 1MEL and 4IDL nanobody structures. FEP, Rosetta, and ThermoMPNN each contained 435 and 409 processed rows for the two templates. Rosetta-scored generated variants and Rosetta-scored random variants each contained 1000 and 1000 processed rows. The MD-derived contact-persistence source was represented by a single processed trajectory-summary table with 1143 rows. Row counts exclude headers.

For a mutation-effect experiment with requested source-label count n , up to n rows were sampled independently from each of the two template-specific source tables using the run seed. The two template-specific tables were assigned separate task identifiers. For an MD-derived source experiment with requested source-label count n , n rows were sampled from the single processed trajectory-summary table. After sampling, source-label rows were split 80/20 into source-training and source-validation rows using the same run seed. These source-validation rows were retained for monitoring and control analyses. In the reported final protocol, however, the validation data passed to the trainer for checkpoint selection and early stopping were filtered to the experimental Tm validation rows.

FEP mutation free-energy labels

The FEP source labels were mutation free-energy changes for variants of the 1MEL and 4IDL nanobody templates. FEP calculations followed the same protocol as our related nanobody thermostability study. Briefly, alchemical mutation simulations were run with NAMD 3 using GPU acceleration (17). Input systems were prepared in VMD (18) using a customized AlaScan workflow (19). Initial structures were obtained from AlphaFold 3 predictions (20). Systems used the CHARMM36 protein force field (21), TIP3P water (22), and 150 mM NaCl.

Each alchemical transformation used a dual-topology mutation setup with a hybrid-residue topology for CHARMM36. The transformation was divided into 20 equally spaced λ windows from wild type to mutant. Forward and backward transformations were simulated, with 1 ns per direction, a 2-fs integration timestep, and 200 ps equilibration per window before data collection.

Simulations were performed in the NPT ensemble at 300 K and 1 atm with Langevin temperature control and a Langevin piston barostat. Bonds involving hydrogen atoms were constrained with SHAKE (23), and long-range electrostatics were evaluated with particle mesh Ewald (24). Relative free energies were estimated with the Bennett acceptance ratio (25). The processed learning labels used the sign convention and [0,1] scaling stored in the FEP CSV files.

Rosetta and ThermoMPNN mutation-effect labels

Rosetta mutation-effect labels were generated with the Rosetta ddg_monomer application (26). For each template, the PDB structure was cleaned to the target chain, variant sequences were compared with the corresponding wild-type sequence, and the sequence differences were converted to Rosetta mutation-list files. The calculation used full-atom Rosetta with the ref2015 score function (27), 50 iterations, Cartesian minimization enabled, local-only optimization disabled, side-chain-minimization-only disabled, minimum-score aggregation, and no PDB dumping. The same Rosetta protocol was used for the template mutation-effect tables, the ESM2-generated variants, and the random-variant control.

For the generated-variant source, ESM2 650M was used to sample 100,000 two-mutation variants from each wild-type template sequence, variants were ranked by pseudolog likelihood, and the top 1% were retained for Rosetta scoring (8). The random-variant control used approximately 1000 two-mutation variants generated by randomly choosing mutation positions and amino acid substitutions, followed by the same Rosetta scoring workflow.

ThermoMPNN source labels were computed from the same template-specific variant sets using ThermoMPNN, a stability-change predictor trained by transfer learning from larger protein-stability data (28). The ThermoMPNN outputs were converted to the same processed CSV format as the other mutation-effect source tasks.

Neural-network architecture

All models used an ESM2 protein language-model encoder (8) followed by lightweight regression heads. Unless otherwise stated, the encoder was facebook/esm2_t6_8M_UR50D. Sequences were tokenized with truncation and padding to a maximum length of 160 residues. The first-token encoder representation was passed to a shared multilayer perceptron with hidden dimensions 256, 128, and 32, ReLU activations, and dropout. A scalar Tm head predicted the scaled target Tm label. In source-transfer models, additional scalar source heads predicted the source-task labels from the same shared representation.

The Tm-only baseline used the ESM2 encoder, the shared trunk, and the Tm head with only experimental

Tm labels. Source-transfer models used the same target path plus source-task heads. The main FEP model used separate source heads for the two template-specific mutation-effect tasks, which was the best-performing source-head design in the source-head comparison. Additional controls evaluated a shared mutation-effect head, a shared head conditioned on a template embedding, and a shared head with template-specific affine calibration. Architecture controls also evaluated residual and latent source-fusion variants, but the main result used the original shared-trunk architecture.

Updated-encoder and frozen-encoder training

Two encoder-training regimes were evaluated. In the updated-encoder regime, the ESM2 encoder and regression heads were optimized jointly. The encoder used a separate learning rate from the newly initialized trunk and heads, so that the pretrained representation was updated more conservatively than the output layers. In the final FEP setting, the selected encoder learning rate was 3×10^{-5} and the head learning rate was the default 3×10^{-4} .

In the frozen-encoder control, all ESM2 parameters were fixed by setting their gradients off. The encoder was run once to precompute sequence embeddings for the target and source examples, and training then optimized only the downstream trunk and task heads. This control asks whether the source labels help when the pretrained sequence representation is used purely as a fixed feature extractor. Encoder-size controls repeated selected Tm-only and FEP-assisted settings with 35M- and 650M-parameter ESM2 encoders.

Loss function and multi-task objective

Each training example was represented as (x_i, y_i, t_i) , where x_i is the sequence, y_i is the scaled regression label, and t_i is the task identifier. The target Tm task used one task identifier, and each source task used its own identifier. A minibatch could contain target and source examples, but each example contributed loss only through the head for its task.

The per-task regression loss was Huber loss with $\delta = 1.0$ on scaled labels. The default multi-task objective used learnable task-uncertainty weights (29). For task k with loss L_k and learned log-scale s_k , the contribution was

$$\frac{L_k}{2\exp(2s_k)} + s_k.$$

The total loss was the sum over the tasks present in the minibatch. Fixed source-task weights were also evaluated as controls, particularly for the MD-derived source labels.

Training, early stopping, and hyperparameter selection

Models were trained with AdamW, cosine learning-rate scheduling, batch size 16, weight decay 0.04, 100 warmup steps, and at most 400 epochs. The default head learning

rate was 3×10^{-4} , the default updated-encoder learning rate was 1×10^{-4} , and the default dropout rate was 0.15. Training used mixed precision on CUDA devices. Checkpoints were saved and evaluated once per epoch.

Within each run, the best checkpoint was the checkpoint with the lowest mean squared error on the experimental Tm validation examples. Early stopping used the same validation metric with a patience of 15 epochs. In source-transfer models, source-label rows were present in the training set, but they were removed from the validation data passed to the trainer for checkpoint selection and early stopping. This is the implementation-level step that ensures that the reported checkpoints are selected by target-task validation performance rather than by source-task performance.

Hyperparameter selection was also performed on the experimental Tm validation split. Candidate settings varied encoder learning rate, head learning rate, dropout, loss-weighting mode, source-head design, and selected architecture controls. The search included encoder learning rates of 3×10^{-5} , 1×10^{-4} , and 3×10^{-4} ; a lower head learning rate of 1×10^{-4} paired with encoder learning rate 3×10^{-5} ; dropout rates of 0.05, 0.15, and 0.30; and fixed source-task loss weights where relevant. Each candidate setting was evaluated as a three-seed ensemble on the Tm validation split. The selected setting for each source class was then evaluated on the held-out test split as a ten-seed ensemble.

Independent run seeds controlled model initialization, target-training subsampling in the target-label scaling experiment, source-label sampling, and source-task train/validation splitting. Final comparisons used seeds 1–10. The label-count curves used fixed source-specific training recipes while varying the number of source labels, rather than retuning a separate model for every label-count point.

Final evaluation and statistics

Final evaluation used the same 396 held-out Tm test examples for all conditions. For each condition, the ten independently seeded models produced scaled Tm predictions for every test sequence. Predictions were averaged in scaled-label space and then inverse-transformed to degrees Celsius. The primary metric was mean absolute error (MAE) in degrees Celsius. Root mean squared error, coefficient of determination, Spearman correlation, and Pearson correlation were also computed, but MAE was used for the main comparisons because it is directly interpretable as average temperature error.

Confidence intervals were estimated by nonparametric bootstrap resampling over held-out test proteins. For a single condition, the absolute-error vector from the ten-seed ensemble was resampled with replacement 10,000 times using a fixed random seed. The reported 90% confidence interval is the 5th to 95th percentile range of the bootstrapped MAE values.

For paired comparisons, the same bootstrap indices were

applied to the absolute-error vectors from the candidate and reference conditions. The candidate-minus-reference MAE difference was then computed for each bootstrap sample. This paired bootstrap preserves the matched-test-example structure of the comparison. In figures and result tables, ΔMAE is candidate minus reference, so negative values indicate lower error than the reference. The one-sided tail probability stored in the result JSON is the fraction of paired bootstrap samples in which candidate-minus-reference ΔMAE is greater than zero. The manuscript emphasizes effect sizes and 90% confidence intervals.

Implementation details

The training and evaluation pipeline is implemented in `prepare.py` and `train.py`. The data loader constructs target and source tasks, applies target-label scaling, performs seeded source-label sampling, filters validation data for target-task checkpoint selection, evaluates seed ensembles, and writes bootstrap statistics. Each result directory contains a machine-readable `scaling.json` file with command-line arguments, resolved hyperparameters, per-point metrics, bootstrap intervals, and paired comparisons.

Results

A target-centered framework for using simulation labels in Tm prediction

We first established an evaluation framework for asking whether simulation-derived labels improve experimental Tm prediction. The target task was nanobody Tm regression from amino-acid sequence. Computational labels were treated as auxiliary prediction tasks rather than as substitutes for experimental Tm. Each model used a pretrained protein sequence model, ESM2, to convert a sequence into a learned representation. The representation was shared by a Tm prediction head and a computed-label head, allowing the computational task to influence the representation while keeping the final objective anchored to experimental Tm prediction (Fig. 1).

The comparison was deliberately target-centered. All candidate settings were selected using an experimental Tm validation set, and the final numbers were computed only on a held-out experimental Tm test set. Improvements were assessed by paired differences in absolute error on the same test examples. This design addresses a common ambiguity in simulation-to-experiment learning: a computational task can be learnable and physically meaningful without improving the target experimental phenotype.

Scaling depends on the physical label

We next varied the number of experimental and computational labels to determine whether more source data translated into better Tm prediction. As expected,

increasing the number of experimental Tm labels improved the model trained only on Tm measurements (Fig. 2). Mutation free-energy labels from FEP calculations also showed a beneficial trend: the largest FEP label-count setting tested gave the best held-out Tm error among the FEP settings. The curve was not strictly monotonic, so we do not interpret it as a clean power law. Instead, we interpret it as evidence that additional mutation free-energy labels can improve the target task when they are selected and evaluated through the experimental Tm endpoint.

The trajectory-summary label behaved differently. This label was derived from MD simulations and measured how strongly native contacts persisted during high-temperature trajectories. Increasing the number of these labels did not yield a robust improvement over learning from Tm measurements alone, even though the computed labels were also generated from simulations. Thus, the label-count experiment already suggested the central result of the paper: the number of computed labels is not sufficient. The transferable physical content of the label matters.

Mutation free-energy labels provide the strongest final improvement

We then compared the selected settings for each computational label. Training on the experimental Tm labels alone achieved a held-out test MAE of 6.61°C. Adding FEP mutation free-energy labels reduced the MAE to 6.26°C, corresponding to a paired improvement of 0.35°C (90% confidence interval, 0.24–0.46°C; Fig. 3). This was the clearest improvement observed across all computational labels.

The other sources showed weaker or negative effects. Structure-based scores from Rosetta assigned to variants proposed by ESM2 improved over learning from Tm labels alone (paired improvement, 0.17°C; 90% confidence interval, 0.03–0.31°C), and ThermoMPNN stability-change labels showed a similar but smaller gain. Rosetta mutation scores and Rosetta-scored random variants were weaker and had confidence intervals that crossed zero. The MD contact-persistence summary worsened the final test error relative to learning from Tm labels alone. These results support a selective, source-dependent interpretation: mutation free-energy-like labels were useful, but a structural trajectory summary was not sufficient for this target task.

The FEP gain is not explained by sequence-model size alone

Because the size of a protein language model can affect downstream performance, we tested whether the benefit of FEP labels could be reproduced simply by using a larger sequence encoder. It could not. In the experimental-only setting, larger 35M and 650M encoders did not improve held-out performance relative to the 8M encoder. In the FEP setting, the 35M and

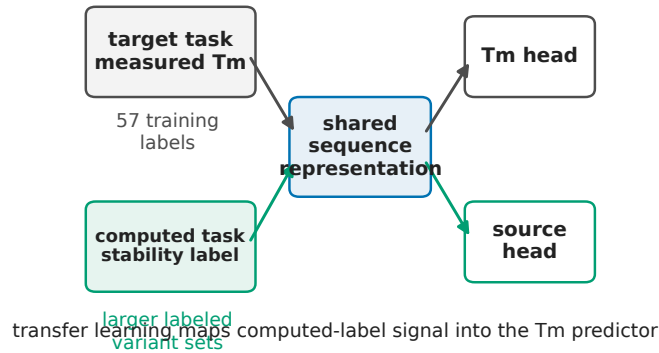
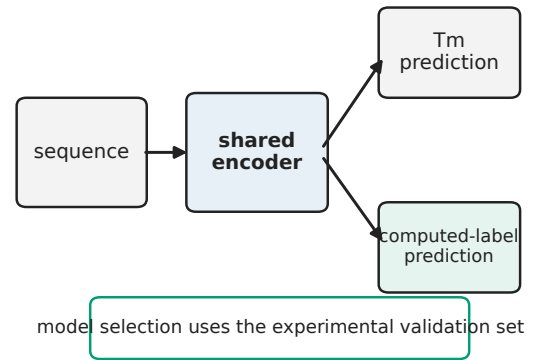
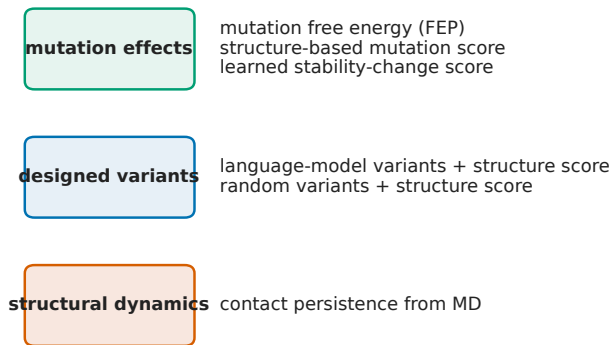
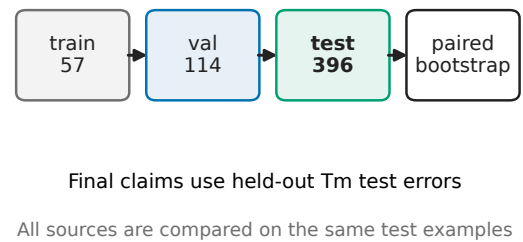
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Fig. 1. Multi-task transfer-learning framework for using simulation data in experimental Tm prediction. (a) Experimental Tm labels are scarce, whereas simulation and physics-based calculations can provide larger computed-label pools. (b) Nanobody Tm prediction is used as a concrete case in which the computed labels are related to, but not identical with, the experimental target. (c) The model shares a pretrained protein sequence representation between the Tm target head and a computed-label head. (d) Model settings are selected on an experimental validation set, and final claims are evaluated on held-out experimental Tm test examples using paired error statistics.

650M encoders reached 6.50 and 6.65°C MAE, respectively, both worse than the 8M FEP model. FEP labels remained beneficial within each size comparison, but the best absolute performance was obtained with the smallest encoder and FEP labels. Thus, in this low-Tm-data regime, the dominant gain came from the physical source label rather than from scaling the language model.

Encoder adaptation helps exploit the computed-label signal

We also asked whether the FEP gain required updating the sequence encoder. In a frozen-encoder control, overall errors increased, but FEP still improved over the frozen experimental-only baseline: the frozen experimental-only model achieved 7.38°C MAE, whereas frozen FEP achieved 7.20°C MAE. The improvement was smaller than in the updated-encoder setting, and the frozen FEP model remained worse than the updated-encoder FEP model. These results indicate

that FEP labels carry useful information even when the representation is fixed, but limited encoder adaptation is important for extracting the strongest target-task benefit.

Physics-labeled generated variants connect transfer learning to design loops

Rosetta labels on variants proposed by the protein language model improved Tm prediction over the experimental-only baseline, whereas the comparison between language-model-proposed and random Rosetta-scored variants was suggestive but not conclusive. The paired improvement of language-model-proposed variants over experimental-only learning was 0.17°C (90% confidence interval, 0.03–0.31°C). By contrast, the paired difference between language-model-proposed and random variants was 0.07°C in favor of language-model-proposed variants, with a confidence interval crossing zero. We therefore interpret this result conservatively.

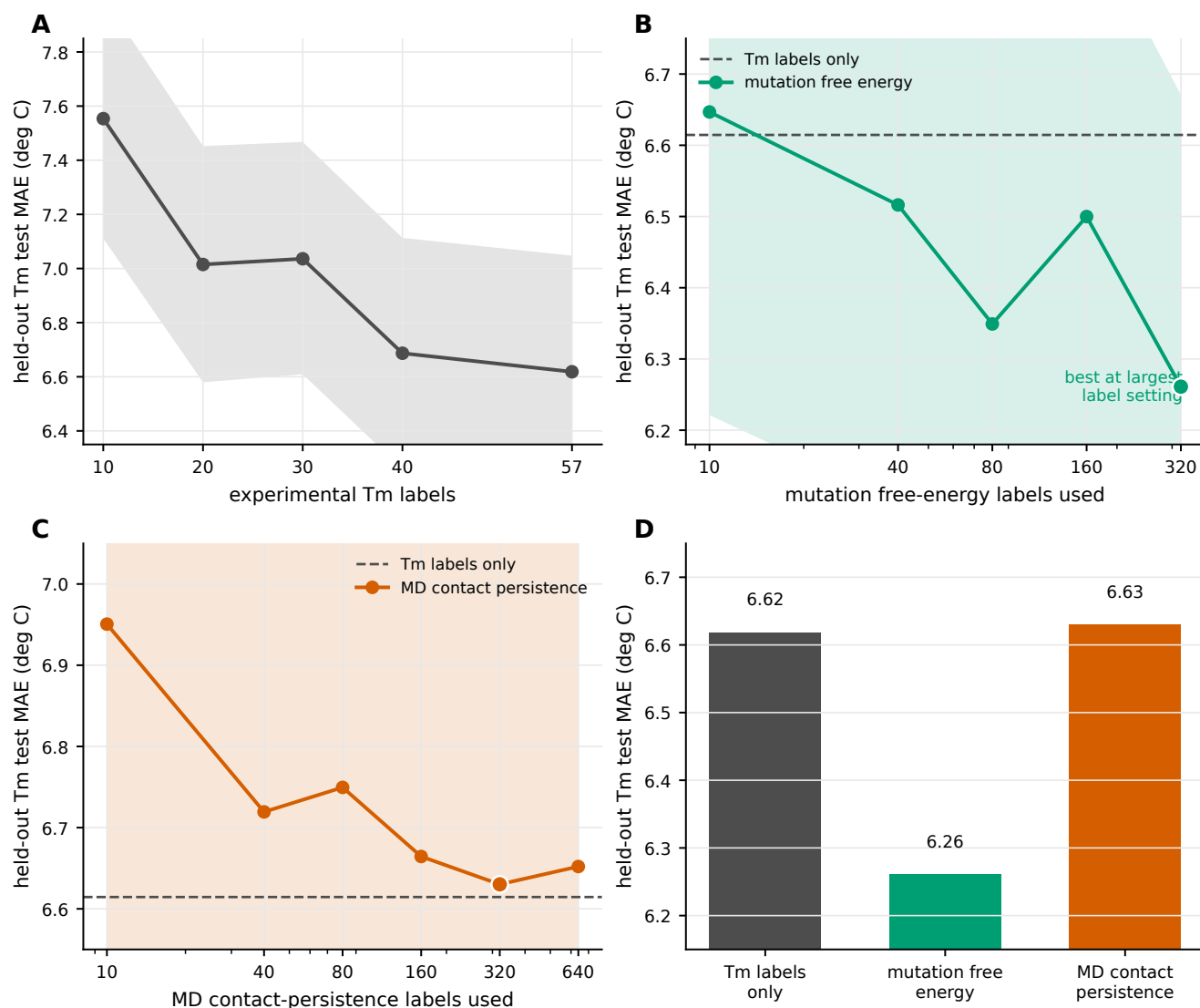


Fig. 2. Label-count scaling of experimental and computational supervision. (a) Learning from experimental Tm labels alone improves as more Tm labels are used. (b) Mutation free-energy labels from FEP reach their best held-out Tm error at the largest computed-label setting tested, although the curve is not strictly monotonic. (c) The MD contact-persistence summary does not show a robust improvement over learning from Tm labels alone. (d) Best points from the label-count sweeps separate FEP from the experimental-only and MD contact-persistence references.

Physics-labeled generated variants retain useful supervised signal for Tm prediction, but the present data do not prove that the current generator is superior to random variant sampling.

Mutation free-energy labels provide a physical interpretation

The final pattern suggests why FEP labels transfer to Tm prediction even though they are not Tm measurements. Experimental Tm labels provide sparse anchors for the absolute thermal-stability phenotype. Mutation free-energy labels provide local information about how stability changes under sequence perturbations. When learned jointly, these two views can shape a representation that is more useful for held-out Tm prediction than either sparse Tm labels alone or a generic simulation-derived summary. The negative result for the MD-

derived contact-persistence label defines the boundary of this interpretation: a simulation label must encode the right physical relation to the target phenotype.

Discussion

Source-label transfer is selective

This study asked how simulation-derived labels should be used when the experimental target is scarce and the simulated quantity is related to, but not identical with, that target. Using nanobody Tm prediction as a case study, we found that FEP mutation free-energy labels improved held-out experimental Tm prediction more clearly than any other source label tested. The result was not reproduced by a simple MD-derived contact-persistence label, nor was it explained by using a larger ESM2 encoder. These findings support a selective view

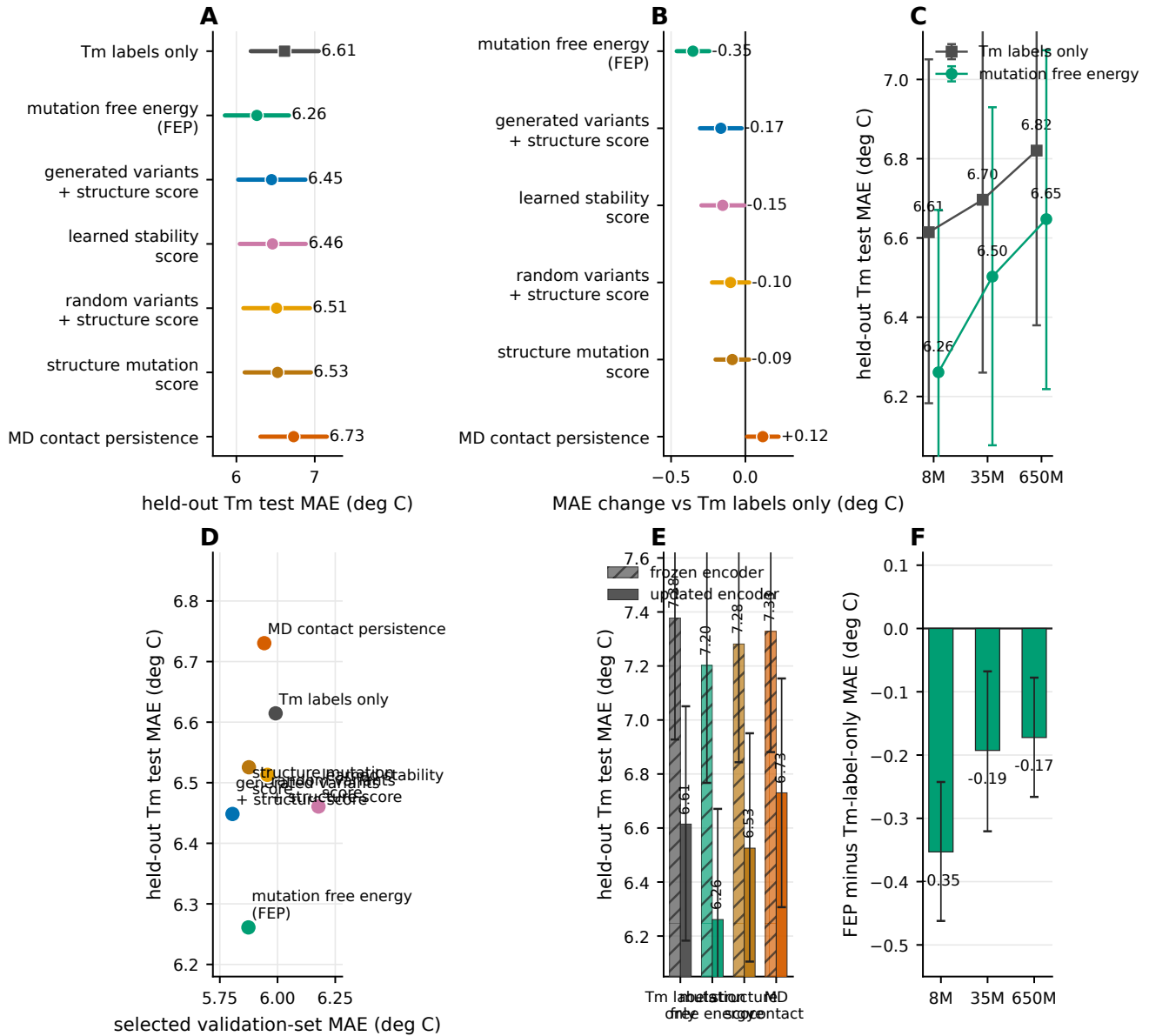


Fig. 3. Final held-out Tm performance across computational labels and controls. (a) Final test MAE for training on experimental Tm labels alone and for each computational label. Intervals show bootstrap confidence intervals over test-set absolute errors. (b) Paired Δ MAE relative to training on Tm labels alone. Negative values indicate improvement. (c) Protein sequence model size controls show that larger encoders do not improve the selected experimental-only or FEP settings. (d) Validation-set performance and held-out test performance are related but not identical, emphasizing the need for final target-task testing. (e) Frozen- and updated-encoder comparisons show that FEP helps in both regimes, with the strongest absolute performance obtained when the sequence representation is updated. (f) Paired FEP gains remain negative across protein sequence-model sizes, but the best absolute error is obtained with the 8M encoder.

of Sim2Real learning for biomolecules: simulation data are useful when the source label encodes physical information that transfers to the experimental phenotype.

Mutation free-energy labels connect relative and absolute stability

The main conceptual point is that the helpful source label did not measure Tm itself. FEP estimates mutation free-energy changes, whereas Tm is an experimental phenotype reflecting the temperature at which folded and unfolded states are balanced under a particular assay condition. These are not interchangeable quantities.

Nevertheless, both describe the same underlying stability landscape from different perspectives. Sparse Tm measurements anchor absolute thermal-stability values for observed sequences, while mutation free-energy labels provide local directions in sequence space. The improvement from FEP therefore suggests that source labels can help even when they are not direct surrogates for the target, provided that they describe a physically relevant perturbation of the target property.

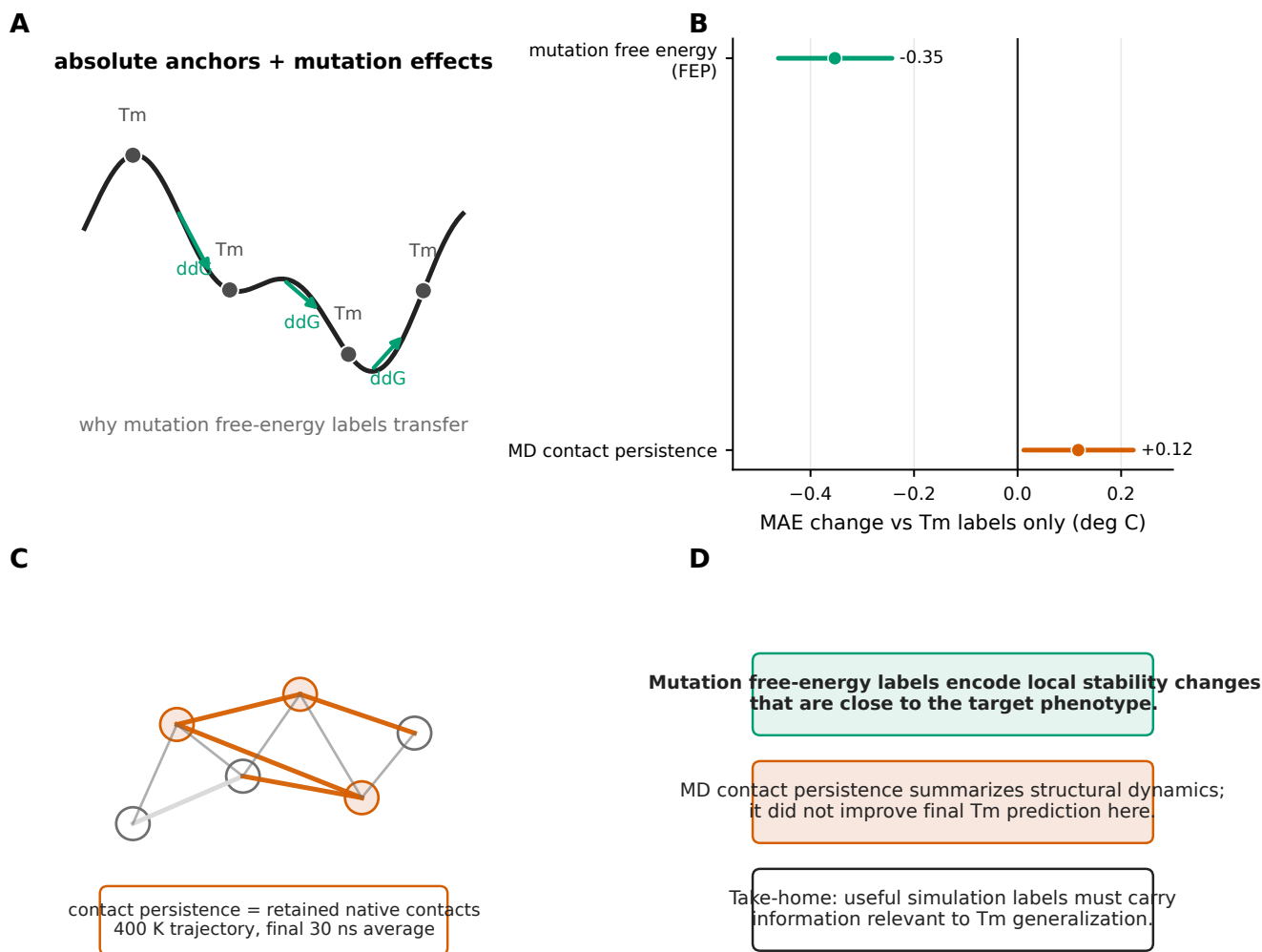


Fig. 4. Interpretation and boundary condition. (a) Experimental Tm labels provide sparse absolute stability anchors, whereas mutation free-energy labels provide local stability-change directions. (b) FEP mutation free-energy labels improve Tm prediction, whereas the molecular-dynamics contact-persistence label does not. (c) Contact persistence summarizes retained native contacts in high-temperature trajectories, but this structural-dynamics signal does not necessarily transfer to held-out Tm prediction. (d) The useful signal is not generic simulation augmentation, but computed-label information with transferable physical content.

Scaling behavior depends on the source label

Model scale is not the main bottleneck

This result extends the materials-informatics Sim2Real argument to a more heterogeneous biomolecular setting. In materials applications, transfer learning often links computational and experimental estimates of related material properties, and scaling behavior can quantify the downstream value of a growing computational database (3). In the present protein case, the most useful computational labels were not additional Tm measurements but mutation free energies. The label-count sweep for FEP was consistent with a beneficial scaling trend but was not smooth enough to claim a universal biomolecular scaling law. The stronger conclusion is source dependence: simulation-derived data should be evaluated by target-task performance, and source labels must be chosen for transferable physical content rather than availability alone.

The encoder controls sharpen this conclusion. Prior work on PLMs has shown that fine-tuning can be critical for downstream protein-property prediction (16), and related nanobody Tm analyses suggest that larger pretrained ESM2 models do not automatically produce better Tm predictors. Our own size controls reached the same practical conclusion: 35M and 650M ESM2 encoders did not outperform the 8M encoder in either Tm-only or FEP-assisted settings. The best model used the 8M encoder with FEP source labels. Thus, for this dataset, the bottleneck was not simply language-model capacity. The more important ingredient was additional supervision carrying information about sequence-dependent stability changes.

Encoder adaptation helps but does not make every source useful

The frozen-encoder experiment further indicates how the source-label signal is used. FEP labels improved performance even when the encoder was fixed, showing that part of the signal can be captured by the existing sequence representation. However, the best FEP model required updating the encoder. This suggests that FEP labels help most when they are allowed to reshape the representation toward stability-relevant sequence directions while still being selected by experimental T_m performance. In practical terms, the result argues for controlled encoder adaptation rather than either fully fixed feature extraction or unconstrained source-task optimization.

The negative result for the MD-derived contact-persistence label is equally important. It shows that the failure mode is not a lack of simulation in general, but a mismatch between the source label and the target phenotype. Contact persistence in high-temperature trajectories may report aspects of structural durability, but it is a compressed trajectory statistic and may miss the mutation-specific energetic effects that determine how sequence changes alter the stability landscape. This distinction matters for future database design. A simulation campaign optimized for target transfer should not only maximize the number of labels; it should choose labels whose physical meaning is expected to transfer to the experimental decision being made.

Generated variants point toward design workflows

The generated-variant result points toward a possible design workflow. Rosetta scores assigned to ESM2-generated variants improved T_m prediction relative to T_m-only learning, although generated variants were not conclusively better than random variants in the current comparison. This should be viewed as a bridge, not a completed design loop. The positive signal indicates that proposed sequences can be labeled by physics-based calculations and used as supervised source data for a T_m predictor. A natural next step is an iterative loop in which a generator proposes variants, physics-based calculations annotate them, the T_m predictor is updated, and candidates are selected for experimental testing. The present work supplies evidence for the supervised learning component of that loop, but it does not test active learning, reinforcement learning, or experimental validation of newly proposed high-T_m nanobodies.

Limitations

Several limitations should guide interpretation. First, the experimental T_m training set is small, and the conclusions should be tested on additional nanobody collections and other protein families. Second, the source-label comparison is limited to the computational labels available here. Other free-energy protocols, stability predictors, trajectory summaries, or structure-based de-

scriptors may transfer differently. Third, the FEP curve does not establish a clean power-law scaling relation; it supports a beneficial trend and a strong final improvement, but not a universal scaling exponent. Fourth, no new experimental measurements were performed to validate high-stability candidates. Finally, raw simulation trajectories may be too large to deposit fully, so reproducibility will depend on processed data, scripts, and clear reporting of the simulation protocol.

Outlook

For low-data experimental protein-property prediction, simulation labels should be treated as design choices rather than as generic data augmentation. The relevant question is not whether simulations can produce many labels, but whether those labels encode information that improves the target task under a target-centered validation protocol. In this study, mutation free-energy labels provided that information for nanobody T_m prediction. The broader strategy is therefore to use experimental data to define the target phenotype, use simulation to annotate physically meaningful perturbations around that phenotype, and evaluate the combined model only on held-out experimental data. This gives a disciplined path from simulation-assisted prediction toward future closed-loop protein engineering.

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AUTHOR CONTRIBUTIONS

Taihei Murakami and Kentaro Sasaki contributed equally to this work. Yasuhiro Matsunaga conceived and supervised the study, acquired funding, developed the analysis and training code, performed the analyses, interpreted the results, prepared the figures, and wrote the manuscript. Kentaro Sasaki performed computational calculations. Taihei Murakami contributed to draft writing. All authors reviewed and approved the final manuscript.

Declaration of competing interest

Taihei Murakami is an employee of Epsilon Molecular Engineering, Inc. Yasuhiro Matsunaga is a scientific advisor of Epsilon Molecular Engineering, Inc.

Data availability

Processed data required to reproduce the main and supplementary figures will be deposited in Zenodo at [ZENODO_DATA_DOI_PLACEHOLDER](#). Raw simulation trajectories are large and will be made available at [RAW_DATA_LOCATION_OR_REQUEST_POLICY_PLACEHOLDER](#).

Code availability

Analysis, training, and figure-generation code will be available on GitHub at [GITHUB_REPOSITORY_](#)

[URL_PLACEHOLDER](#). An archival release will be deposited in Zenodo at [ZENODO_CODE_DOI_PLACEHOLDER](#).

AI-use disclosure

AI-assisted tools were used to support language editing, code development, and figure preparation. The authors reviewed and take responsibility for all manuscript content, analyses, and conclusions.

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Supplementary Information

Supplementary Note 1: Experimental target split and target-label scaling

All reported comparisons used the same experimental Tm endpoint from NbBench and the same held-out test split. The processed experimental split contained 57 training sequences, 114 validation sequences, and 396 test sequences. The target CSV files used the columns text for amino-acid sequence and label for Tm in degrees Celsius. The training split was used to fit neural-network parameters. The validation split was used for checkpoint selection, early stopping, and selection among candidate model settings. The test split was not used until those settings had been fixed.

The Tm regression target was scaled before training and converted back to degrees Celsius before evaluation. The scaling transform was fit only on the 57 target-training labels. In code, the transform is a RobustScaler followed by a MinMaxScaler with range [0,1]. The fitted transform was then applied to validation labels during training and inverted for validation and test predictions. The validation and test labels therefore did not influence the fitted target-label transform. In experiments that varied the amount of experimental Tm supervision, only the 57-row training split was subsampled; the validation and test splits remained fixed.

Supplementary Note 2: Processed source-label tables and task identifiers

Table 1 summarizes the processed source-label data used by the reported comparisons. Row counts exclude header rows and were verified from the processed CSV files. Mutation-effect sources were represented by two template-specific source tasks, corresponding to the 1MEL and 4IDL nanobody templates. These files used seq as the sequence column and ddg_scaled01 as the [0,1]-scaled source-label column. The MD-derived contact-persistence source was represented by a single processed trajectory-summary table with sequence, raw trajectory-summary quantities, and a ddg_scaled01 training label.

Table 1. Processed target and source-label data.

Data type	Role	Processed rows	Use in reported comparisons
Experimental Tm	Target train	57	Model fitting
Experimental Tm	Target validation	114	Checkpoint and setting selection
Experimental Tm	Target test	396	Final held-out evaluation
FEP mutation free energy	Source tasks	435 + 409	Main source task
Rosetta mutation score	Source tasks	435 + 409	Source comparison
ThermoMPNN mutation score	Source tasks	435 + 409	Source comparison
Rosetta-scored random variants	Source tasks	1000 + 1000	Source comparison
Rosetta-scored generated variants	Source tasks	1000 + 1000	Source comparison
MD-derived contact persistence	Source task	1143	Source comparison and scaling control

The implementation assigns task identifiers as follows: experimental Tm, 0; the two template-specific mutation-effect tasks, 1 and 2; the primary MD-derived source task, 3; and an optional second MD-derived source task, 4. The reported main comparisons used the Tm task, one computational source class at a time, and the task identifiers needed for that source. The source-task identifiers determine which regression head and which loss term are applied to each minibatch example.

Supplementary Note 3: Seeded sampling, tokenization, and validation construction

Each independent run uses the run number as the seed. The seed is applied to Python random, NumPy, PyTorch, and CUDA when CUDA is available. It controls neural-network initialization, experimental Tm subsampling in the target-label scaling experiment, source-label sampling, and the source-task train/validation split. Final comparisons used seeds 1–10.

For mutation-effect experiments with requested source-label count n , up to n rows were sampled independently from each of the two template-specific source tables using the run seed. Thus, the final FEP setting with $n = 320$ provided 320 sampled 1MEL rows and 320 sampled 4IDL rows before the source split. For MD-derived contact-persistence experiments, n rows were sampled from the single processed trajectory-summary table. Sampled source rows were split 80/20 into source-training and source-validation rows. Source-validation rows were kept in the result files for monitoring and control analyses, but the production training scripts filtered the validation data passed to the trainer so that checkpoint selection and early stopping used experimental Tm validation rows.

Sequences were tokenized with the ESM2 tokenizer using truncation and padding to MAX_LENGTH = 160 residues. Tokenization used fixed-length padding so that target and source examples entered the same model interface. The

frozen-encoder controls precomputed the first-token ESM2 embedding for each example and stored it as an embedding field before head training. Updated-encoder models kept token IDs and attention masks and updated the ESM2 weights during training.

Supplementary Note 4: Neural-network implementation

The default base encoder was facebook/esm2_t6_8M_UR50D. The first-token ESM2 representation was passed through a three-layer trunk with dimensions hidden-size $\rightarrow 256 \rightarrow 128 \rightarrow 32$, ReLU activations, and dropout. The main architecture used one shared trunk with a Tm regression head and source-specific scalar heads. The same Tm path was used in Tm-only and source-transfer models, so the comparison changes the source-task supervision rather than replacing the target model.

Architecture controls implemented three additional ways to let source-task representations influence the Tm path. The residual variant adds a gated source-informed correction to the Tm prediction. The latent variant predicts a correction from the concatenation of Tm features, source features, and their element-wise product. The mixture variant learns a gate between a Tm-only expert and a source-informed expert. These controls were used to test whether the source signal required a stronger fusion architecture. The selected main result did not require these additional fusion paths.

For template-specific mutation-effect labels, the production comparison used separate scalar heads for the two template tasks. Source-head controls also tested a single shared mutation-effect head, a shared head conditioned on a learned template embedding, and a shared base head with template-specific affine calibration. For MD-derived source labels, the primary MD head corresponded to task identifier 3; the code also supports an optional second MD head with task identifier 4, although the main figures use one MD-derived source at a time.

Supplementary Note 5: Training objective, optimizer, and checkpoint selection

Each example contributed Huber loss with $\delta = 1.0$ on its own scaled target and its own task head. The default multi-task objective used learnable task-uncertainty weights. In the alternative fixed-weight setting, Tm and mutation-effect losses used weight 1.0, whereas MD-derived losses used the selected fixed MD weight. Fixed MD weights were included because an MD-derived source label can have a different noise scale and label distribution from the experimental Tm target.

Training used AdamW, weight decay 0.04, a cosine learning-rate schedule, 100 warmup steps, batch size 16, and at most 400 epochs. The default head learning rate was 3×10^{-4} , the default updated-encoder learning rate was 1×10^{-4} , and the default dropout rate was 0.15. Updated-encoder models used separate optimizer parameter groups for ESM2 parameters and newly initialized trunk/head parameters. Frozen-encoder controls set all ESM2 parameters to non-trainable and optimized only the downstream network.

Checkpoints were evaluated and saved once per epoch. Within each run, the saved model used for prediction was the checkpoint with the lowest mean squared error on experimental Tm validation examples. Early stopping monitored the same validation metric with patience 15 and threshold 0. The per-run checkpoint directory contains model.safetensors, training_args.bin, and hyperparameters.json. The latter records the resolved architecture, encoder, loss-weighting, dropout, and learning-rate settings used for that run.

Supplementary Note 6: Candidate-setting searches

The source comparison used a two-stage procedure. First, each candidate setting was evaluated as a three-seed ensemble on the experimental Tm validation split. Second, the selected setting for each source class was rerun or collected as a ten-seed ensemble and evaluated on the held-out test split. The same procedure was applied to the Tm-only reference and to each computational source class.

The common candidate grid included the default setting; encoder learning rates 3×10^{-5} , 1×10^{-4} , and 3×10^{-4} ; a lower head learning rate of 1×10^{-4} paired with encoder learning rate 3×10^{-5} ; and dropout rates 0.05, 0.15, and 0.30. For source classes where loss balance could plausibly change the conclusion, the grid also included fixed source-loss weights. Tm-only candidates included shared, residual, and latent architecture variants. Mutation-effect source candidates used the shared trunk for the main source comparison, with a separate source-head search for template-head design. MD-derived source candidates used the residual architecture in the main source search because preliminary validation runs found that this family was the strongest MD-derived-source candidate.

The script scripts/source_screen.py generated the main source comparison and the selected-source test summaries. The script scripts/ddg_head_search.py generated the mutation-effect head controls. The script scripts/abcd_search.py generated the target-only, mutation-effect-only, MD-derived-only, and combined-source controls. The low-level training/evaluation entry point was prepare.py, which calls train.py.

Supplementary Note 7: Machine-readable outputs and figure provenance

Each call to `prepare.py` writes a structured `scaling.json` file in the result directory. This file records the experiment name, timestamp, short Git commit, command-line arguments, relevant environment variables, resolved encoder mode, hyperparameters, per-label-count MAE values, bootstrap confidence intervals, per-example absolute errors, and paired bootstrap summaries. Because the per-example absolute errors are stored, paired comparisons can be recomputed without rerunning neural-network training.

The main source comparison is summarized by `results/source_screen/final_source_screen_summary.json`. The frozen-encoder source comparison used in the control panels is summarized by `results/source_screen/final_frozen_core_summary.json`. The source-head controls are summarized by `results/ddg_head_search/final_ddg_head_summary.json`. Additional source-combination controls are summarized by `results/abcd_search/final_abcd_summary.json`. The main figure set is assembled by `plot/make_outline_figures.py`, which also writes the tabular source summary paper/`tex/figures/outline_figure_source_screen.tsv`.

Supplementary Note 8: Statistical analysis

The primary reported statistic is MAE on the 396 held-out test examples after averaging predictions from ten seeded models. Predictions are averaged in scaled-label space and then inverse-transformed to degrees Celsius before computing errors. Confidence intervals were computed by resampling test examples, not by treating training seeds as independent experimental replicates. For single-condition MAE intervals, 10,000 bootstrap samples were drawn from the absolute-error vector and the 5th and 95th percentiles were reported.

For paired comparisons, both conditions were resampled with the same bootstrap indices and the candidate-minus-reference MAE difference was computed for each resample. This paired procedure preserves the fact that all models are evaluated on the same test proteins. The one-sided bootstrap tail probability recorded in the result JSON is the fraction of paired bootstrap samples in which candidate-minus-reference ΔMAE is greater than zero. It is reported as a diagnostic for the direction of the paired difference, but the main text emphasizes the effect size and the 90% confidence interval.

Supplementary Note 9: Interpretation of the label-count curves

The experimental T_m label-count curve is an internal reference showing that the pipeline detects the expected improvement when more target labels are added. The FEP curve evaluates whether increasing a physically related mutation-free-energy source improves the same held-out T_m endpoint. The FEP curve is beneficial but not strictly monotonic, so it should not be interpreted as evidence for a universal power law. The MD-derived contact-persistence curve shows that increasing the amount of a simulation-derived label is not sufficient by itself. The relevant conclusion is source dependence: target-task transfer depends on whether the source label encodes physical information that is useful for experimental T_m prediction.